Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/13/2019
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Development and Testing of Biomarkers to Determine Individual Astronaut Vulnerabilities to Behavioral Health Disruptions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2014	**End Date:**	01/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 1/31/2020 per NSSC information (Ed., 9/13/19) NOTE: Extended to 7/31/2019 per NSSC information (Ed., 8/7/18) NOTE: Extended to 7/31/2018 per K. Ohnesorge/JSC (Ed., 7/14/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17) NOTE: End date shows as 7/31/2017 per NSSC information (Ed., 6/2/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barger, Laura Ph.D. (Brigham & Women's Hospital) Czeisler, Charles M.D., Ph.D. (Brigham and Women's Hospital/Harvard Medical Center) Duffy, Jeanne Ph.D. (Brigham & Women's Hospital) Kristal, Bruce Ph.D. (Brigham & Women's Hospital)		
Grant/Contract No.:	NNX14AK53G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	It is well established that sleep deficiency and circadian desynchrony impair alertness and performance. It is also increasingly recognized that the response to sleep loss and circadian desynchrony is highly individualized and both challenge- and task-dependent. This high degree of inter-individual variability makes it very difficult to predict who will be at risk and when they will be at risk, and therefore a 'one-size-fits-all' approach is likely to be suboptimal. What is needed are biomarkers of individual vulnerabilities and resiliencies to sleep loss and circadian desynchronization that can be used to predict problems and intervene when necessary. In this project, we will first identify candidate biomarkers in a retrospective analysis from neurocognitive and psychological measures collected over the past 17 years in our laboratory analog. These, and previously untested biomarkers, will then be validated in new limited laboratory analog studies to inform a new space analog study that will test the feasibility and utility of a set of core biomarkers to predict and assess variation in neurocognitive and psychological function associated with living in Antarctica during a 12-month mission. These innovative studies have a high potential to identify reliable biomarkers that are suitable for operational use.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of a candidate biomarker panel that delineates inter-individual differences in sleep and circadian rhythms has enormous potential for the development of individualized predictive models of vulnerabilities and resiliencies to sleep deprivation and circadian misalignment for a wide range of occupational settings, for example, in shift-workers and other unusual environments (military, shipping, mining, etc).
Task Progress:	There are three main phases to our project. In Phase 1, we will identify candidate biomarkers to predict neurocognitive and psychological responses to sleep deprivation and circadian misalignment from a retrospective analysis of existing sleep, circadian, and performance data of more than 1,500 subjects that have been studied over the past 17 years. In Phase 2 of this project, we will conduct a series of short, targeted laboratory studies to test prospectively the predictive power of selected biomarkers and to measure novel biomarkers absent from our original data sets in order to inform operational analog studies testing the predictive value of these biomarkers in operational conditions. Our innovative goal is to generate sets of biomarker panels from sleep, performance, circadian, lipidomic and/or metabolomic biomarker profiles to identify interindividual differences in performance under: 1) baseline conditions; 2) acute sleep deprivation; and 3) circadian misalignment. In Phase 3, we will assess candidate biomarkers in Antarctic expeditioners just prior to or immediately at the start of a 6-12 month overwintering period to predict individual vulnerabilities to sleep deprivation and circadian misalignment based on our findings from Phases 1 and 2. We will then collect from these individuals a number of subjective and objective measures of individual behavioral health outcomes, including sleep, circadian rhythms, urinary cortisol and melatonin levels and a range of neurocognitive performance outcomes and alertness, to validate our predictions. We have completed Phase 1 of the project, namely the development of a preliminary candidate biomarker panel that can predict an individual's response to sleep loss phenotype (i.e., resilient, intermediate, or vulnerable, as defined by performance at 20 hours of wakefulness) from neurobehavioral performance collected in the individual at baseline after a habitual (8 hour) night of sleep 2 days earlier. We have compiled and analyzed retrospective performance and alertness data from over 160 subjects (over 1400 days of intensive physiological monitoring) who were studied under identical laboratory conditions. The first goal of our Phase 1 analysis was to develop a metric to define response to sleep loss phenotype within an individual. Prior definitions of response to sleep loss have been based almost exclusively on performance on the psychomotor vigilance task (PVT). Thus, we developed other metrics incorporating data from several neurobehavioral tests, including the Karolinska Sleepiness Scale (KSS), non-numeric bipolar Visual Analog Scales (VAS), auditory PVT (aPVT), 4-minute addition calculation test (ADD), 1.5-minute digit symbol substitution test (DSST), and 10-minute auditory and visual versions of the PVT, to determine whether data from other neurobehavioral performance tasks provide additional information about response types. We explored four metrics for defining the response to sleep loss phenotype in these individuals: a metric based on visual PVT lapses only (metric 1); a metric derived from principal component analysis of 74 variables from all 6 neurobehavioral tests (metric 2); a metric derived from principal component analysis of 8 representative variables from all 6 neurobehavioral tests (metric 3); and a metric derived from principal component analysis of 12 representative variables from all neurobehavioral tests excluding the two versions of the PVT (metric 4). Comparison of the metrics showed that similar information is obtained from the three metrics that include data from the PVT (metrics 1-3) compared with data from other neurobehavioral tasks (metric 4). Therefore, we determined that metric 1, using lapses on the visual PVT, was sufficient for defining response to sleep loss phenotype. The final definitions of each phenotype are as follows: resilient (fewer than 6 lapses), intermediate (between 6 and 30 lapses), and vulnerable (more than 30 lapses). This metric can discriminate ~50% of our subject population (studied under inpatient laboratory conditions) as non-resilient to sleep loss (i.e., intermediate or vulnerable response to sleep loss) at ~20 h awake from one neurobehavioral performance assessment during rested conditions (i.e., at ~8 hours of wake after a habitual 8-hour night of sleep) with ~100% accuracy. Our findings suggest that these individuals who are poor performers during rested conditions exhibit non-sleepiness-related performance impairment that is exacerbated by subsequent sleep loss. Among the remaining ~50% of our subject population, who are high-level performers during rested conditions, the distribution of response to sleep loss phenotype is expected to be approximately equal across resilient, intermediate, and vulnerable phenotypes. A model developed to discriminate response to sleep loss phenotype at 20 h awake in these high-level performers from a single neurobehavioral performance assessment 2 days earlier during rested conditions has an overall accuracy of ~64%, suggesting that neurobehavioral performance alone is insufficient to discriminate response to sleep loss phenotype in these individuals. Our metric that discriminates high-level from poor performers during rested conditions has been implemented as a screening tool to select participants for our 6-day Phase 2 study; only high-level performers have been included as participants in our study. These high-level performers who also met all our other inclusion/exclusion criteria have been enrolled in our 6-day laboratory-based study for Phase 2 of this project. In this laboratory study, we have collected multiple biomarker samples during baseline and sleep deprivation to determine whether there are one or more "omics"-based measures that provide a greater sensitivity and specificity to identify individuals who are resilient versus

non-resilient to sleepiness-related performance impairment. We started recruiting for this study in January 2016, and have consented and screened 60 potential candidates in total. Forty-five were excluded due to not meeting study inclusion/exclusion criteria, including 8 participants who were identified as poor performers from the 35 participants who completed our initial screening test battery. We have completed studies on 15 individuals. Enrollment in this study is now complete.

We have conducted analyses of lipidomic and metabolomic biomarker profiles in a subset of participants who have completed the study. Specifically, lipidomic analysis of blood samples collected every 4 hours during both baseline (days 2-3) and constant routine (days 4-5) has been completed in 14 subjects (6 females). Lipids were extracted from human plasma using dichloromethane:methanol. The lipid extracts were then analyzed by reversed-phase liquid chromatography-mass spectrometry (LC-MS). Full-scan MS analysis was done on a stand-alone orbitrap mass spectrometer (Exactive, Thermo Scientific) in both the positive and negative ionization modes. Chromatographic alignment and framing to select features was performed using SIEVE 2.0. Features that represented lipids were then identified using an in-house lipid database. Metabolomic analysis has also been conducted on these blood samples using a CoulArray platform, and we are currently in the process of identifying metabolite peaks for further analysis.

Initial analysis of lipidomics data has identified 340 unique lipids, ~42% of which exhibit a rhythm during rested conditions and ~28% of which exhibit a rhythm during constant routine when subjects were sleep deprived. In the coming year, we will continue to analyze these lipidomic and metabolomic markers at both the group and individual level to determine a candidate biomarker panel that identifies response to sleep loss phenotype.

In preparation for Phase 3, we have compiled the data from over 50 expeditioners who were studied during the Antarctic winter. These subjects were instructed to complete the ANAM-ICE performance battery approximately once per month several times per day for 1-2 days. Our analysis was conducted primarily on 28 subjects from this cohort who had completed a test battery both at baseline (defined as a test session completed during the first month of the overwintering period after a night of at least 7 hours of sleep between the first 4 to 10 hours awake) and after at least 16 hours of wakefulness during a subsequent month of testing. As in Phase 1, we have started to test different metrics from the ANAM-ICE performance battery to define “resilient,” “intermediate,” or “vulnerable” response phenotypes after more than 16 hours of wakefulness. One metric that we have developed is based on performance on the simple reaction time test of the ANAM-ICE battery. We first normalized each subject’s response based on mean and standard deviation of normative data from each subject’s age group [Vincent et al., 2012]. Then we computed the 25th and 75th percentiles of these normalized values across all subjects across all test sessions. Scores in the top 25th percentile were classified as resilient, scores in the bottom 25th percentile were classified as vulnerable, and scores between the 25th and 75th percentile were classified as intermediate. We then compared the response at baseline to responses from test sessions at >16 hours of wakefulness. We found that we were able to discriminate the response phenotype at >16 hours of wakefulness from baseline performance with an overall accuracy of ~65%. We are continuing to develop this model by incorporating information from other neurobehavioral tests at baseline to improve its overall accuracy. We plan to prospectively apply this model to baseline data collected from expeditioners who will be overwintering in Antarctica.

Reference:

Vincent, A.S., et al., Automated Neuropsychological Assessment Metrics (v4) Traumatic Brain Injury Battery: military normative data. Military Medicine, 2012. 177(3): p. 256-69.

Bibliography Type: Description: (Last Updated: 11/14/2025)

Abstracts for Journals and Proceedings

St Hilaire MA, Rahman SA, Sullivan JP, Kristal BS, Quackenbush J, Duffy JF, Barger LK, Czeisler CA, Lockley SW. "Development and testing of biomarkers to determine individual astronauts’ vulnerabilities to behavioral health disruptions." 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. Abstracts. 2019 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019